

Linear programming

Background material

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Linear program

Linear program, standard form:

$$\begin{aligned} \min_x \quad & c^T x \\ \text{s.t.} \quad & Ax = b, \\ & x \geq 0, \end{aligned}$$

with $x, c \in \mathbb{R}^n$, $b \in \mathbb{R}^m$, $A \in \mathbb{R}^{m \times n}$.

Assumptions: $\text{rank}(A) = m$, $m \leq n$.

Standard form

Any linear program can be transformed to the standard form. Consider for instance the program

$$\min_x c^T x \text{ s.t. } Ax \geq b.$$

Note that there is no sign constraint on x .

We first add a vector z of surplus variables:

$$\begin{aligned} \min_x \quad & c^T x \\ \text{s.t.} \quad & Ax - z = b, \\ & z \geq 0. \end{aligned}$$

We next decompose x as the difference of two non-negative variables:

$x = x^+ - x^-$, where $x^+ = \max\{x, 0\} \geq 0$, and $x^- = \max\{-x, 0\} \geq 0$.

Standard form (cont'd)

The problem becomes

$$\begin{aligned} \min_x \quad & (c \quad -c \quad 0) \begin{pmatrix} x^+ \\ x^- \\ z \end{pmatrix}, \\ \text{s.t.} \quad & (A \quad -A \quad -I) \begin{pmatrix} x^+ \\ x^- \\ z \end{pmatrix} = b, \begin{pmatrix} x^+ \\ x^- \\ z \end{pmatrix} \geq 0. \end{aligned}$$

The inequalities constraints of the form $x \leq u$ or $Ax \leq b$ can be handled by adding slack variables:

$$\begin{aligned} x \leq u &\Leftrightarrow x + w = u, \quad w \geq 0, \\ Ax \leq b &\Leftrightarrow Ax + y = b, \quad y \geq 0. \end{aligned}$$

Basic solutions

Without loss of generality, suppose that the first m columns are independent, and form

$$\mathbf{A} = (\mathbf{B} \quad \mathbf{D})$$

Basic solution: $\mathbf{x} = (\mathbf{x}_B \ 0)$, with $\mathbf{B}\mathbf{x}_B = \mathbf{b}$.

Degenerate basic solution: some components of $\mathbf{x}_B$ are equal to zero.

Feasible basic solution: basic solution such that $\mathbf{x}_B \geq 0$.

Fundamental theorem of linear programming

Consider an LP under standard form, with $\mathbf{A} \in \mathbb{R}^{m \times n}$, and $\text{rank}(\mathbf{A}) = m$.

- If there exists a feasible solution, then there exists a feasible basic solution.
- If there exists a feasible optimal solution, then there exists a feasible optimal basic solution.

Extreme points and extreme rays

Let $P = \{x \mid Ax = b, x \geq 0\}$ be non-empty.

- The **extreme points** (vertices) of P are exactly its feasible basic solutions. There is therefore a **finite** number of them, at most $\binom{n}{m}$.
- The **recession cone** of P is $\{d \mid Ad = 0, d \geq 0\}$, and it has finitely many **extreme rays**.
- **Resolution theorem**: every $x \in P$ is the sum of a convex combination of the extreme points and of a non-negative combination of the extreme rays.

Consequently a linear program over P either attains its optimum at an extreme point, or is unbounded along an extreme ray r such that $c^T r < 0$.

A polyhedron admits an extreme point as soon as it contains no line, which $x \geq 0$ guarantees here.

Duality: symmetric form

- **Primal** program

$$\begin{aligned} \min_{x \geq 0} \quad & c^T x \\ \text{s.t.} \quad & Ax \geq b \end{aligned} \tag{P}$$

- **Dual** program

$$\begin{aligned} \max_{\lambda \geq 0} \quad & \lambda^T b \\ \text{s.t.} \quad & A^T \lambda \leq c \end{aligned} \tag{D}$$

- $A \in \mathbb{R}^{m \times n}$, $c, x \in \mathbb{R}^n$, $\lambda, b \in \mathbb{R}^m$: x gathers the **primal** variables, λ the **dual** ones.
- Dualizing (D) gives (P) back: the dual of the dual is the primal.

Duality: from the standard form

The program

$$\begin{array}{ll}\min_x & c^T x \\ \text{s.t.} & Ax = b \\ & x \geq 0\end{array}$$

is equivalent to

$$\begin{array}{ll}\min_x & c^T x \\ \text{s.t.} & Ax \geq b \\ & -Ax \geq -b \\ & x \geq 0\end{array}$$

Duality: asymmetric form

The dual problem can then be written as

$$\begin{aligned} \max_{u,v} \quad & u^T b - v^T b \\ \text{s.t.} \quad & u^T A - v^T A \leq c^T \\ & u \geq 0 \\ & v \geq 0 \end{aligned}$$

or, with $\lambda = u - v$,

$$\begin{aligned} \max_{\lambda} \quad & \lambda^T b \\ \text{s.t.} \quad & \lambda^T A \leq c^T \end{aligned}$$

Asymmetric form: $\lambda \in \mathbb{R}^m$ is unrestricted in sign.

Primal-dual conversion

Minimization	Maximization
Constraints $\geq$ $\leq$ $=$	Variables ≥ 0 ≤ 0 unconstrained
Variables ≥ 0 ≤ 0 unconstrained	Constraints $\leq$ $\geq$ $=$

Weak duality

(Asymmetric or symmetric form)

If x and λ are feasible for the primal and the dual, respectively, then

$$c^T x \geq \lambda^T b$$

Proof.

For any primal feasible x and dual feasible λ , we have

$$\lambda^T b \leq \lambda^T A x \leq c^T x.$$



Corollary

If x_0 and λ_0 are feasible for the primal and the dual, respectively, and if

$$c^T x_0 = \lambda_0^T b,$$

then x_0 and λ_0 are optimal for their respective problem.

But we have said nothing about the feasibility of one problem with respect to the other one!

Strong duality

If one of the problems (P) or (D) has a finite optimal solution, the other problem also has a finite optimal solution, and the corresponding values of the objective functions are equal. If one of the problems has an unbounded objective, the other problem has no feasible solution.

If a program is infeasible, this does not imply that its dual is unbounded: it can be infeasible as well. “Bounded” below means that the program is feasible and has a finite optimal value.

Primal / Dual	Bounded	Unbounded	Infeasible
Bounded	possible	impossible	impossible
Unbounded	impossible	impossible	possible
Infeasible	impossible	possible	possible

Polyhedral cones and Farkas' lemma

Positive hull, or cone generated by the columns of A :

$$\text{pos } A = \{z \mid z = Ax, x \geq 0\}.$$

It is exactly the set of right-hand sides z for which $Ax = z, x \geq 0$, admits a solution.

Farkas' lemma

Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. Exactly one of the following holds:

- $\exists x \geq 0$ such that $Ax = b$, i.e. $b \in \text{pos } A$;
- $\exists u \in \mathbb{R}^m$ such that $A^T u \geq 0$ and $b^T u < 0$.

The second vector is a **certificate of infeasibility**: it defines a hyperplane separating b from $\text{pos } A$.

Optimality and duality

The optimality can be deduced from KKT conditions.

In LP, we do not need a constraint qualification to apply KKT conditions.

Lagrangian:

$$\mathcal{L} = c^T x - \pi^T (Ax - b) - s^T x.$$

KKT conditions: Lagrangian vectors π and s s.t.

$$A^T \pi + s = c,$$

$$Ax = b,$$

$$x \geq 0,$$

$$s \geq 0,$$

$$x_i s_i = 0, \quad i = 1, 2, \dots, n \quad (\text{complementarity}).$$

Dual problem

Let (x^*, π^*, s^*) be a vector satisfying the KKT conditions. We have that

$$c^T x^* = (A^T \pi^* + s^*)^T x^* = (Ax^*)^T \pi^* = b^T \pi^*.$$

It is furthermore easy to show that the (necessary) KKT conditions are sufficient.

Dual problem:

$$\max_{\pi} b^T \pi, \text{ s.t. } A^T \pi \leq c,$$

or, equivalently,

$$\max_{\pi} b^T \pi, \text{ s.t. } A^T \pi + s = c, s \geq 0.$$

The value function of a linear program

Consider the optimal value as a function of the right-hand side,

$$z(b) \stackrel{\text{def}}{=} \min_x \{c^T x \mid Ax = b, x \geq 0\},$$

with $z(b) = +\infty$ when the program is infeasible: $\text{dom } z = \text{pos } A$.

By duality, $z(b) = \max_{\pi} \{\pi^T b \mid A^T \pi \leq c\}$, where the feasible set **does not depend on b** . Denoting by $\pi^1, \dots, \pi^r$ its extreme points, for any $b \in \text{dom } z$,

$$z(b) = \max_{i=1, \dots, r} (\pi^i)^T b.$$

- z is **convex and piecewise linear** in b , with finitely many pieces;
- any optimal dual solution π^* is a **subgradient**: $z(b') \geq z(b) + (\pi^*)^T (b' - b)$;
- as a function of the costs c , z is a minimum of linear functions, hence **concave and piecewise linear**.